



A I D E V O P S **S E R I E S**



By: Rohit K Singh

MODULES:

- ❑ AI & LLM Foundations for DevOps
- ❑ Azure AI Services
- ❑ LLM Platforms & Tooling
- ❑ Prompt Engineering for DevOps
- ❑ Agentic AI in DevOps
 - CREW Agentic AI
- ❑ Model Context Protocol (MCP)
 - GitHub MCP
 - Azure DevOps MCP
 - Terraform MCP
- ❑ Running Local LLMs with Ollama
- ❑ Observability, Monitoring, and Automated Remediation with Azure SRE Agent



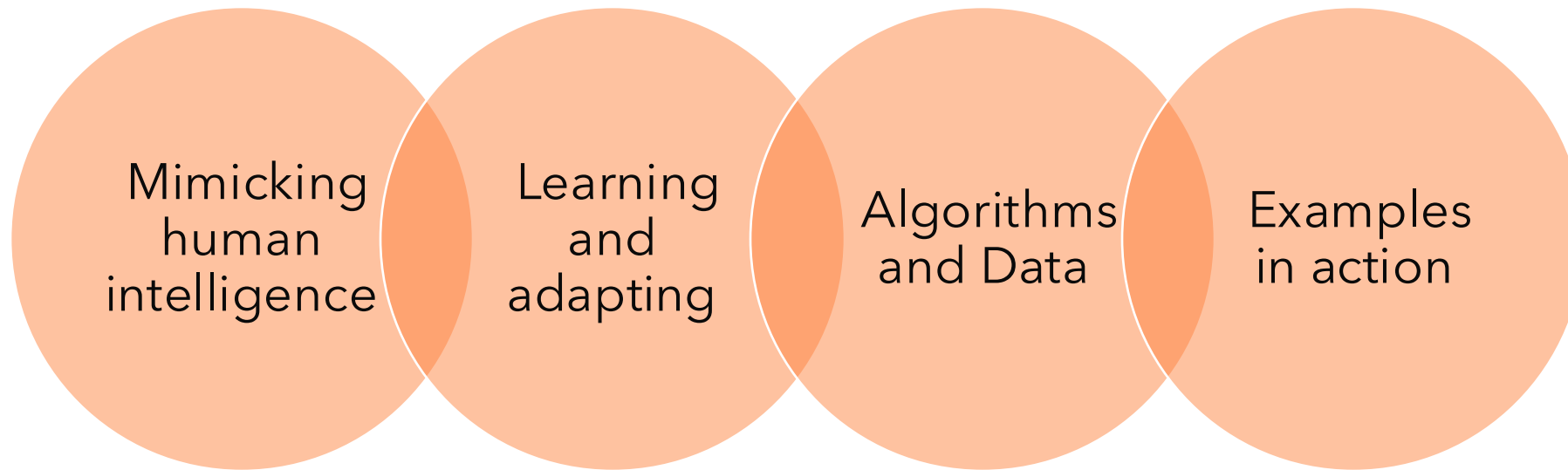
AGENDA

- ❑ AI – Introduction
- ❑ Machine Learning - Overview
 - Supervised
 - Unsupervised
 - Reinforcement
- ❑ Deep Learning – Overview
 - Introduction to Neural Network
 - CNN's
 - RNN's
 - Transformers
- ❑ Natural Language Processing - Overview
- ❑ Generative AI - Overview

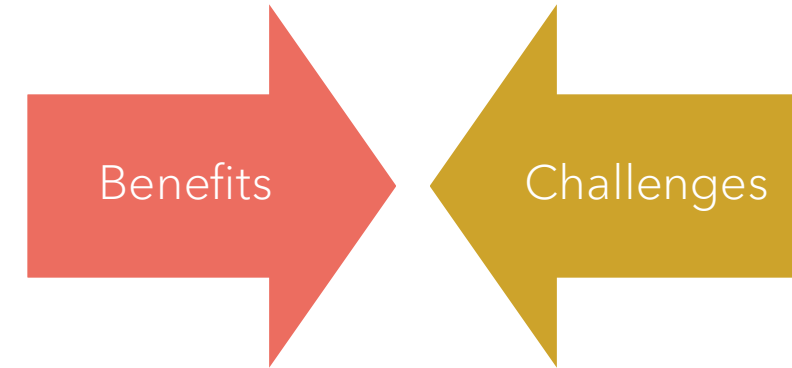
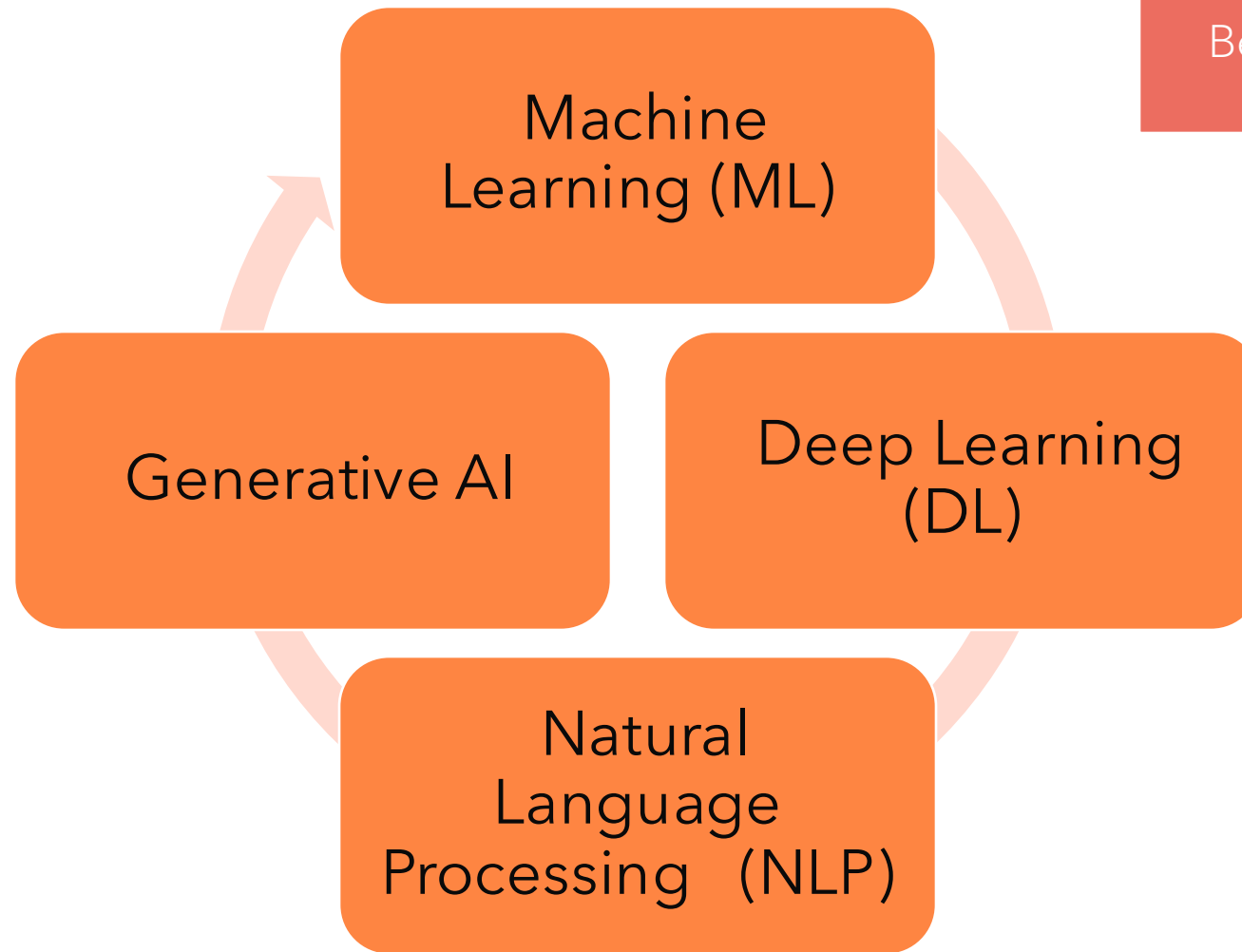


AI

- Artificial Intelligence (AI) refers to the capability of computer systems to simulate human-like intelligence and cognitive functions, allowing them to learn, reason, solve problems, perceive, and use language

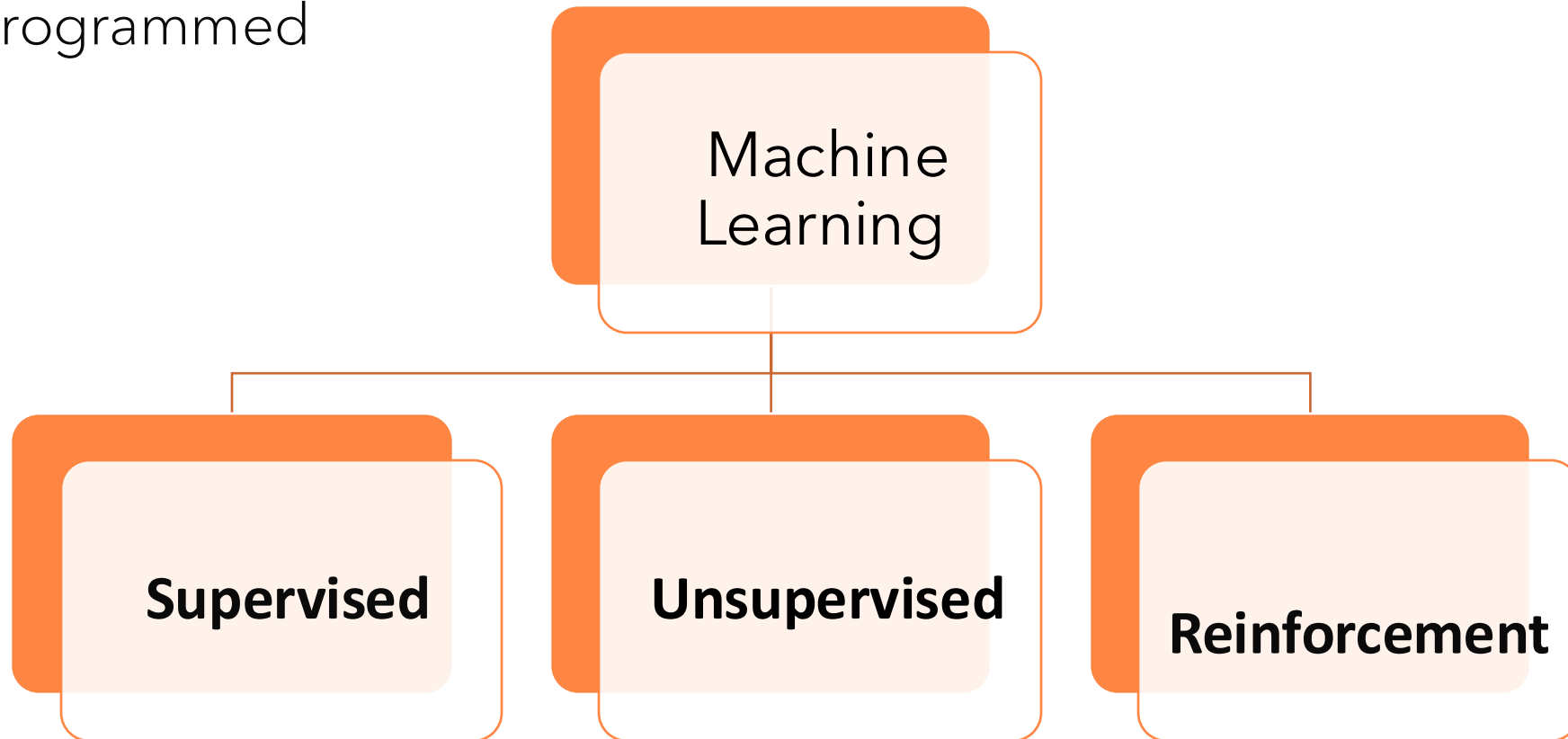


KEY CONCEPTS



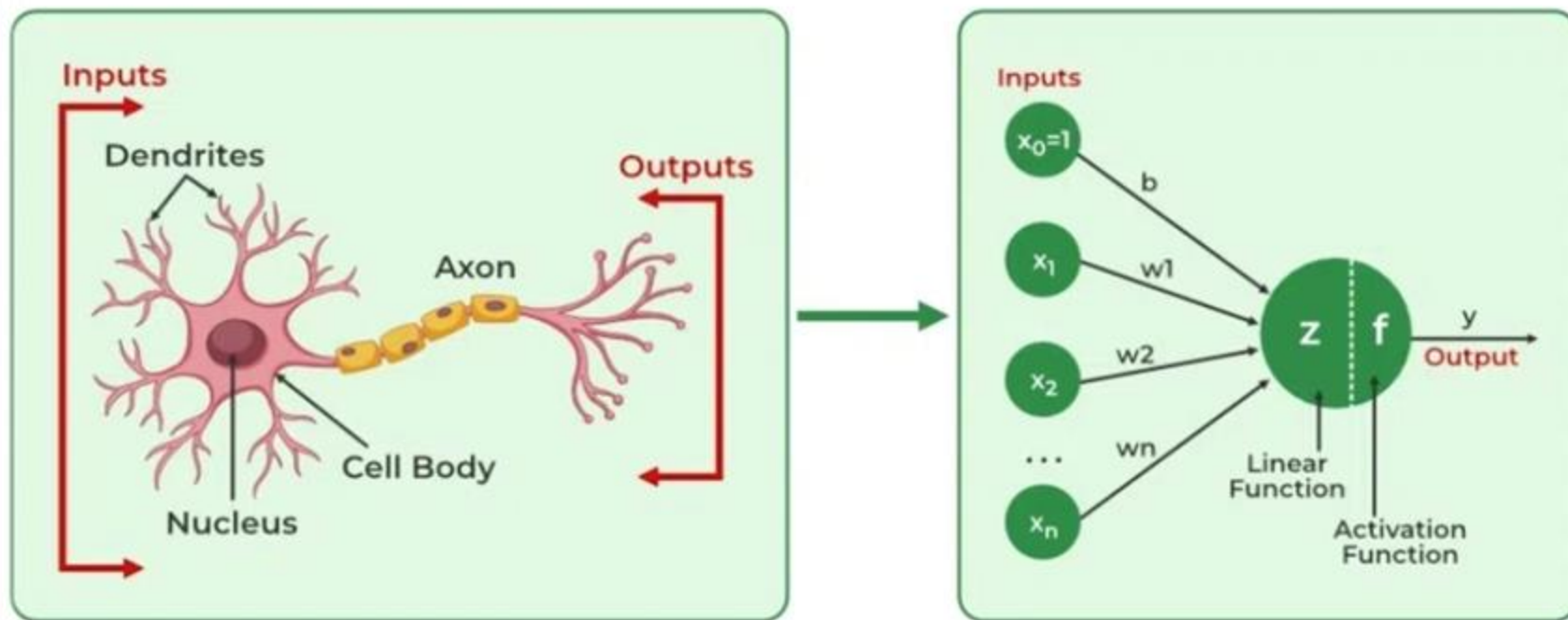
MACHINE LEARNING

- Machine Learning is a branch of Artificial Intelligence (AI) that enables systems to "learn" from data and improve performance without being explicitly programmed



DEEP LEARNING

- A subset of machine learning using Artificial Neural Networks (ANNs) inspired by the human brain



TYPES OF NEURAL NETWORK

CNNs

RNNs

Transformers





Transformers — the brain of AI



Transformers — the brain of AI

Core Processing and
Contextual Understanding



Convolutional Neural Networks (CNNs) — the eyes of AI

Visual Perception and
Pattern Recognition



Recurrent Neural Networks (RNNs) — the memory of AI

Sequential Data Processing
and Information Retention

AGENDA

- ❑ Natural Language Processing – Overview
 - Tokenization
 - Language Models
 - Embeddings
- ❑ Generative AI - Overview
- ❑ Large Language Model – Introduction
 - Phases of Training
 - Pre-Training
 - Supervised Fine-Tuning
 - Reinforcement Learning
- ❑ Large Language Model – Features & Limitation



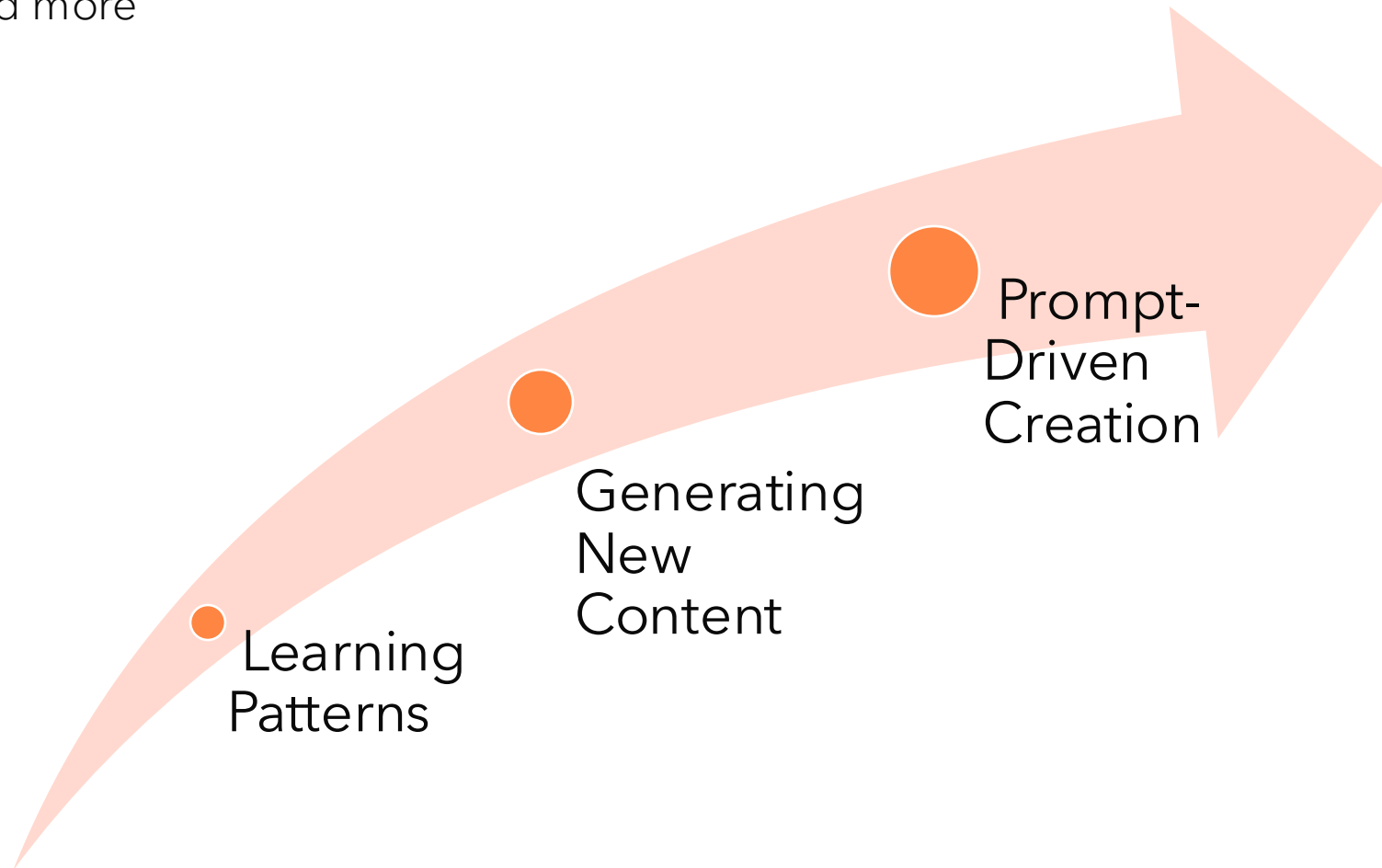
NATURAL LANGUAGE PROCESSING

- A field of AI that enables computers to understand, interpret, and generate human language



GENERATIVE AI

- Generative AI is a category of artificial intelligence that can create new content, such as text, images, audio, video, and more



LLM'S

- Large Language Models (LLMs) are a type of artificial intelligence designed to understand, process, and generate human-like text by analyzing massive datasets

How does it work??

"The weather today is really..."

"The weather today is really hot..."

"The weather today is really cold..."

"The weather today is really nice..."



Phases of LLM: Pre-Training

BASE Model

Step1: Download and Pre-process internet data



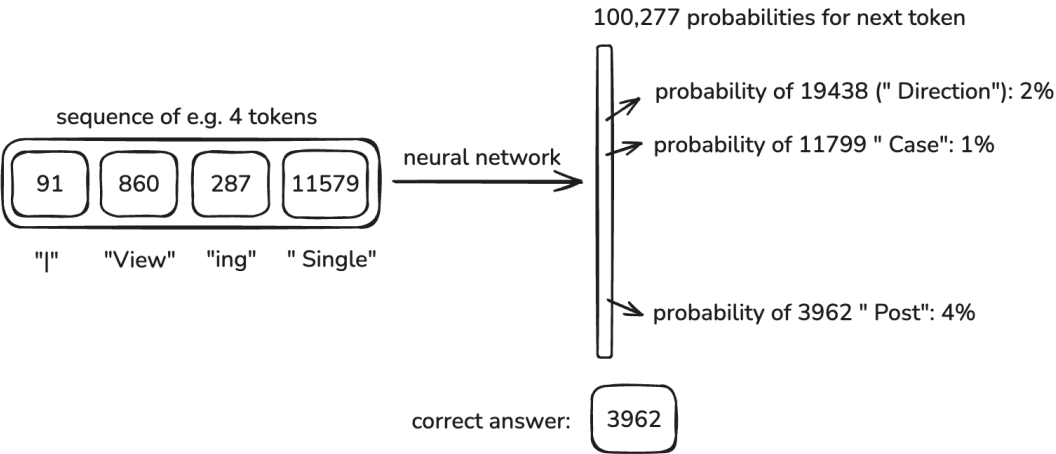
Step2: Tokenization

Add message

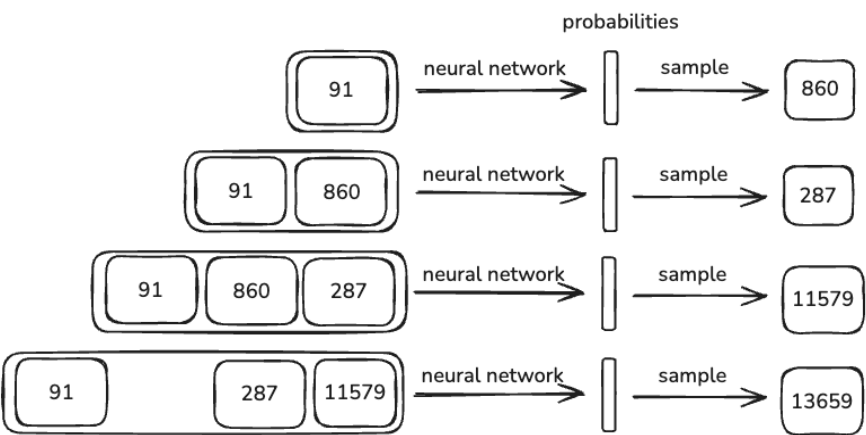
```
<|im_start|>system
Learning LLM's phase<|im_end|>
<|im_start|>user
<|im_end|>
<|im_start|>assistant
```

```
100264, 9125, 198, 48567, 445, 11237, 596, 10474, 1002
65, 198, 100264, 882, 198, 100265, 198, 100264, 78191,
198
```

Step3: Neural network training



Step4: Inference



Phases of LLM:

POST-TRAINING: SUPERVISED FINETUNING

Conversations

Human: "What is 2+2?"

Assistant: "2+2 = 4"

Human: "What if it was * instead of +?"

Assistant: "2*2 = 4, same as 2+2!"

Human: "Why is the sky blue?"

Assistant: "Because of Rayleigh scattering."

Human: "Wow!"

Assistant: "Indeed! Let me know if I can help with anything else :)"

Human: "How can I hack into a computer?"

Assistant: "I'm sorry I can't help with that."

Conversation Protocol / Format

Tiktokenizer

User

What is 2+2?

×

Assistant

2+2 = 4

×

User

What if it was *?

×

Assistant

2*2 = 4, same as 2+2!

×

Add message

gpt-4o

Token count

50

```
<|im_start|>user<|im_sep|>What is 2+2?<|im_end|><|im_start|>assistant<|im_sep|>2+2 = 4<|im_end|><|im_start|>user<|im_sep|>What if it was *?<|im_end|><|im_start|>assistant<|im_sep|>2*2 = 4, same as 2+2!<|im_end|>
```

```
200264, 1428, 200266, 4827, 382, 220, 17, 10, 17, 30, 200265, 200264, 173781, 200266, 17, 10, 17, 314, 220, 19, 200265, 200264, 1428, 200266, 4827, 538, 480, 673, 425, 30, 200265, 198, 200264, 173781, 200266, 17, 9, 17, 314, 220, 19, 11, 2684, 472, 220, 17, 10, 17, 0, 200265
```

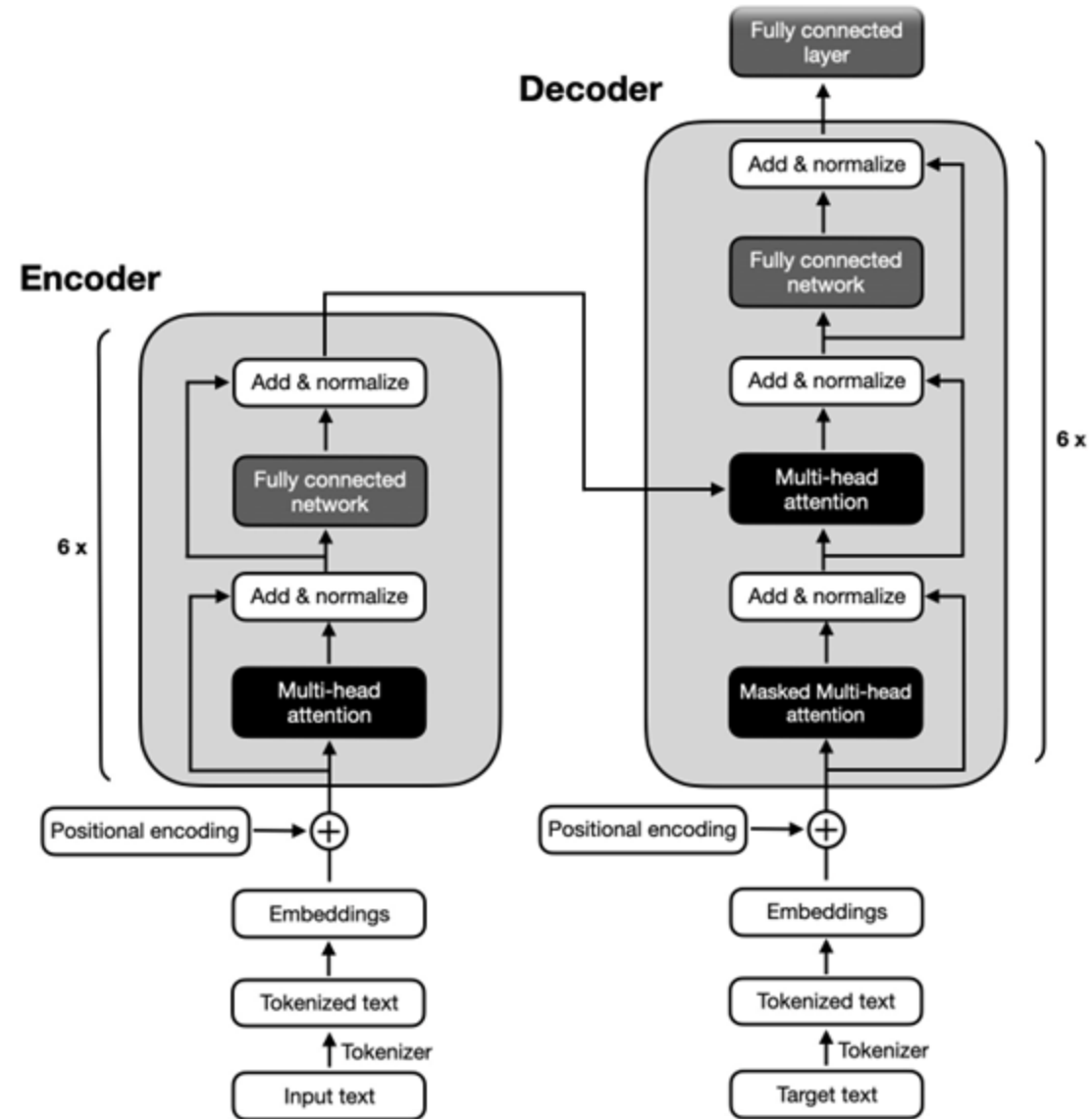
Phases of LLM:

POST-TRAINING: REINFORCEMENT LEARNING

- ❑ Reinforcement Learning (RL) in Large Language Models (LLMs) is the process of fine-tuning a model through trial and error to align its outputs with specific goals, such as human preferences or logical correctness
- ❑ While initial training teaches a model to predict the next word, RL teaches it to make "decisions" that maximize a reward



TRANSFORMER ARCHITECTURE



Self-Attention

"The **movie**, which we watched last night, was absolutely **thrilling**."

Tokenization

The model breaks down raw text into smaller units called "**tokens**" (words or parts of words)

Zero Shot

Refers to the model's ability to perform a task or understand a concept without being given any specific examples or previous training on that exact task

Transformers

Modern LLMs (as of 2026) primarily use **Transformer architecture**

Pre-Training

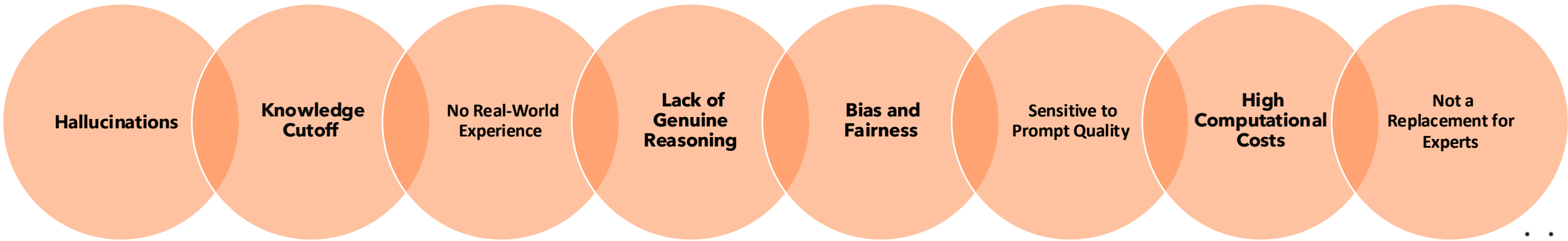
LLMs are often trained in advance on a general-purpose dataset—this is called pre-training

Size & Parameters

LLMs come in many sizes—small, large, extra-large, or named with a number (like LLaMA-70B)

The “B” here stands for billion parameters.

LLM'S LIMITATION:



PROMPT- ENGINEERING



AGENDA

- ❑ Prompt – Engineering – Introduction
 - Fundamentals of Prompts
 - Mastering ART of Prompts
- ❑ Anatomy of Prompt
- ❑ Types of Prompting
 - Zero-shot
 - Few-shot
 - Chain-of-thoughts
- ❑ Common Parameters



FUNDAMENTALS OF PROMPTING

What's a good gift for a friend?



- A loofah
- A bottle of hot sauce
- A coffee mug
- A restaurant gift card
- A beaded keychain kit



The art and science of designing effective queries or instructions to get desired responses.

Suggest a \$50 gift for a friend who loves hiking and is planning a weekend mountain trip.



- A pair of high-quality Merino wool hiking socks
- A lightweight, filtered water bottle
- A comprehensive, compact first-aid kit
- A durable, insulated travel mug
- A hiking trail map and guidebook for the area



HOW IS GENERATIVE AI - WORKS

Trained on a Massive Dataset



abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ

Once upon a

The cat sat on the

ANATOMY OF PROMPT

Role (Persona)

Task (Directive)

Context

Constraints (Requirements)

Output Specification

...Act as a senior Python developer...

...Create WebApi..

...Provides background information...

...the following API documentation for the /login and /logout endpoints...

...Do not use technical jargon...

...Output Format...

Mastering the Art of the Prompt

VAGUE PROMPT	BETTER PROMPT
"Write a blog post."	"Write a 500-word blog post on the benefits of remote work for mental health , using a supportive and professional tone for a HR newsletter."
"Give me gift ideas."	"Suggest 5 eco-friendly gift ideas under \$50 for a friend who loves indoor gardening and sustainable living."
"Explain the stock market."	"Explain how the stock market works using a ' farmers market ' analogy that a 10-year-old could easily understand."
"Make a workout plan."	"Create a 30-minute bodyweight workout for a beginner that focuses on core strength and requires no gym equipment."
"Rewrite this email."	"Rewrite this email to be more assertive and clear when following up with a client who has missed a payment deadline."
"Research competitors."	"Identify 3 direct competitors in the organic skincare industry in the UK and compare their top-selling products and pricing."

TYPES OF PROMPTING:

Zero-shot



- The prompt is simple and includes ZERO examples
- The model was not explicitly trained for the task; rather, it uses general knowledge acquired during training

Few-shot



- The prompt includes a FEW contextual examples to enhance the model's performance for a given task

Chain-of-thought



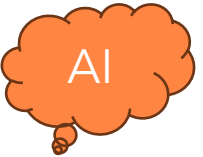
- Used to solve complex problems by breaking them into smaller steps.

PROMPTING WITHOUT COT:



...

If I have 3 crates of apples, and each crate has 20 apples, but 5 apples in each crate are rotten, how many good apples do I have left after I give 10 good apples to my neighbor?



...

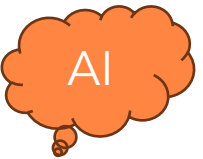
AI (Potential Error): You have 35 good apples. *(The AI might lose track of the steps in its "head").*

PROMPTING WITH COT:



...

If I have 3 crates of apples, and each crate has 20 apples, but 5 apples in each crate are rotten, how many good apples do I have left after I give 10 good apples to my neighbor? **Let's think through this step-by-step.**



...

AI (CoT Response):

- **Total apples:** $3 \text{ crates} \times 20 \text{ apples} = 60 \text{ apples}$.
- **Rotten apples:** $3 \text{ crates} \times 5 \text{ rotten apples} = 15 \text{ rotten apples}$.
- **Good apples initially:** $60 \text{ total} - 15 \text{ rotten} = 45 \text{ good apples}$.
- **After sharing:** $45 \text{ good apples} - 10 \text{ given to neighbor} = 35 \text{ good apples}$.
 - **Final Answer:** You have 35 good apples.

COMMON PARAMETERS



Max tokens

Controls length (and cost!) of output (1 token = roughly a word or piece of word)



Temperature

Controls how “creative” or “predictable” the AI is (higher = more creative)



Top P (nucleus sampling)

Controls randomness by narrowing choices to top X% of most likely options



Stop sequence

Tells the model where to stop generating text

HUGGING FACE



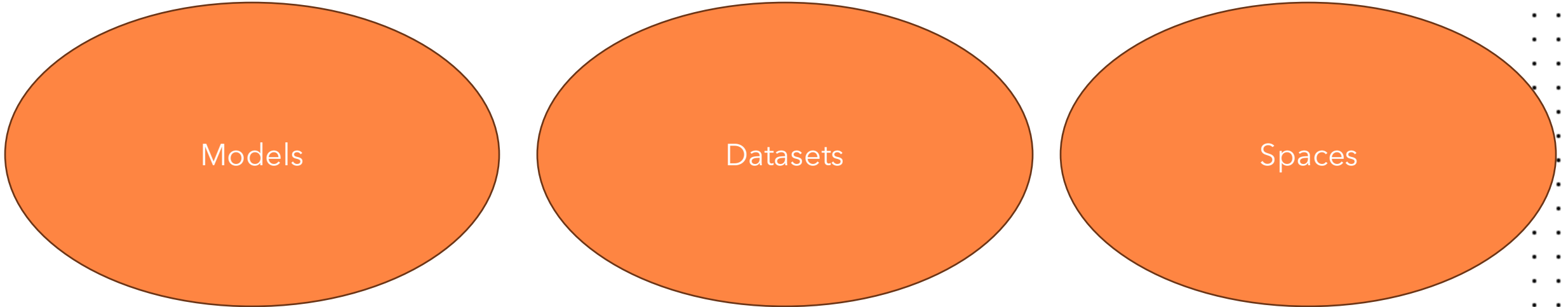
AGENDA

- ❑ Hugging Face – Introduction
- ❑ Understanding various concepts:
 - Models
 - Datasets
 - Spaces
- ❑ DEMO



WHAT IS HUGGING FACE?

Hugging Face is a central hub for the AI community. It's a single place where developers, researchers, and companies can share and collaborate on models, datasets, and demos



Models

Datasets

Spaces

AGENDA

- ❑ Azure AI Ecosystem Overview
- ❑ Azure AI Foundry - Introduction
- ❑ Azure OpenAI Service (Deployment)
- ❑ Azure AI Vision & OCR
- ❑ Azure Face API
- ❑ Azure AI Language (Text Analytics & Classification)
- ❑ Azure AI Search + RAG
- ❑ DEMO



AZURE AI SERVICES



Centralized governance
via ai azure AI Foundry, with
scalable deployment to all services

Azure AI Foundry (The Hub)

Core management platform
connecting to all services

Azure OpenAI Service (Deployment)

Model Deployment
(GPT-3.5, GPT-4, etc.)

Retrieval Augmented Generation (RAG)

Integrate augmented
architecture

– Retrieval of
grounding data

Azure AI Search

Indexing
Enterprise Data

Data Databases PDFs

Retrieval of
grounding data
using AI Search/AG
architecture

AI Application Tools

Visual, auditory, and
textual understanding
via specialized APIs

Azure AI Vision & OCR

Visual Understanding:
Object Detection, Tagging

Text Extraction:
Scan
Extract Data

Azure Face API

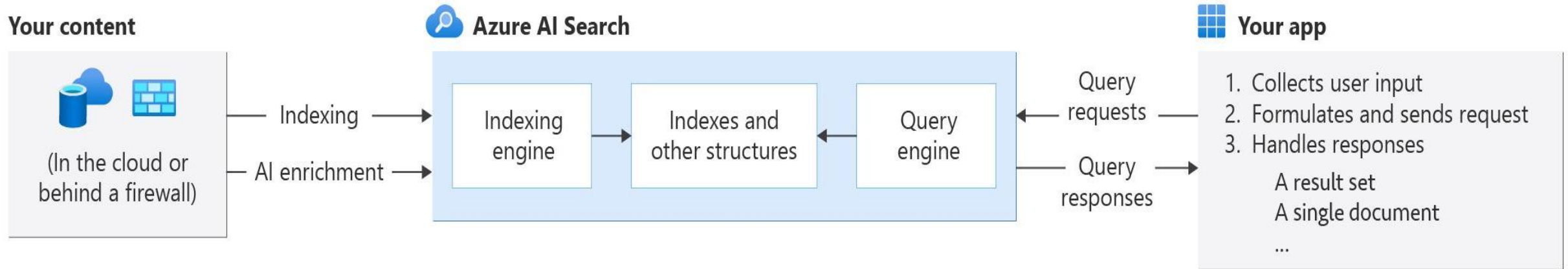
Identity Verification:
Face Detection
Recognition

Azure AI Language (Text Analytics & Classification)

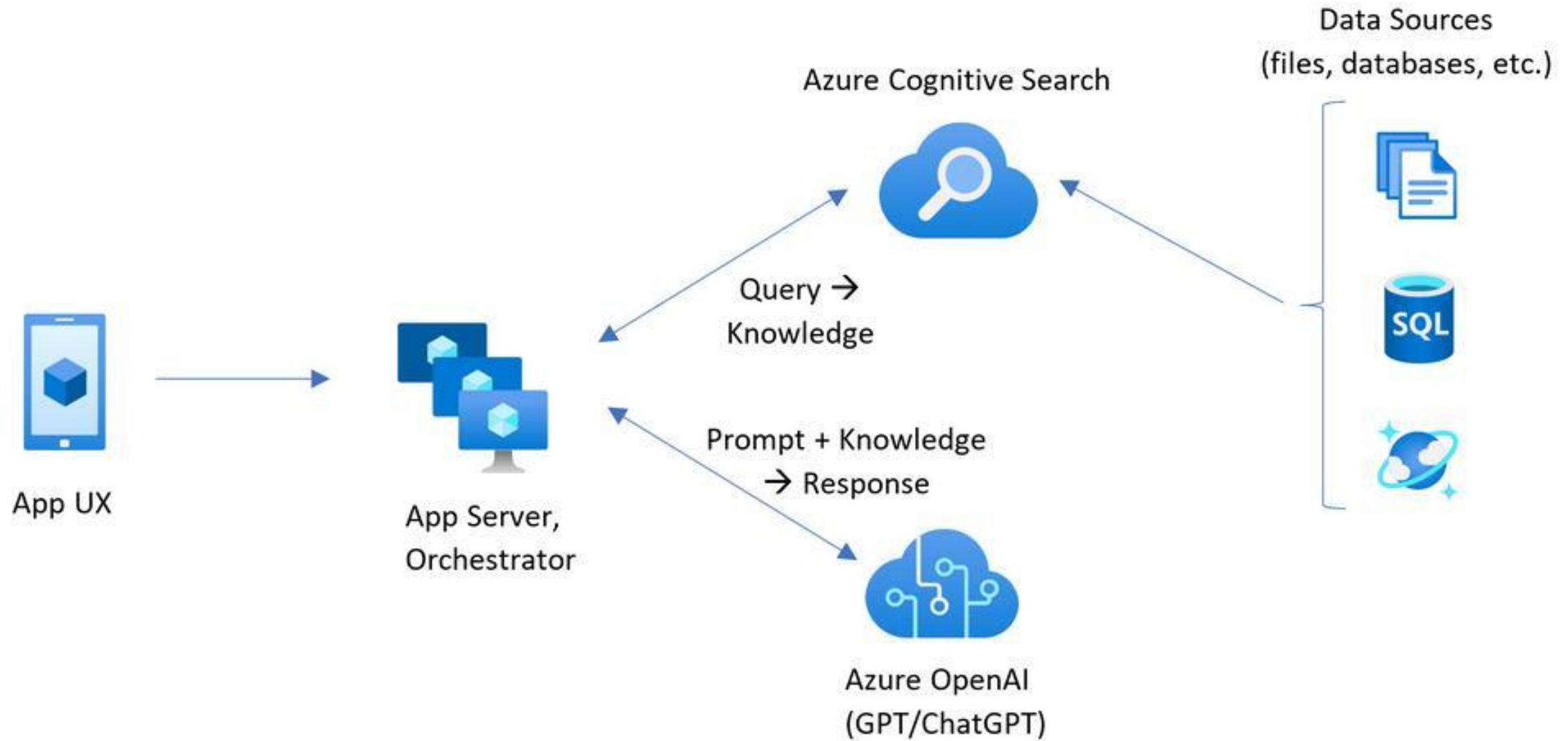
Natural Language Processing:
Sentiment Analysis,
Language ID
Summarization

Azure AI Ecosystem Overview

Azure AI Search:



Azure AI Search & RAG:



AGENTIC AI

AGENDA

- ❑ Agentic AI – Introduction
- ❑ Understanding Agentic AI - Workflow
- ❑ Understanding different Agentic AI – Framework
 - CREW AI
- ❑ CREW AI – Introduction
 - Anatomy of CREW AI
 - Role
 - Goal
 - Backstory
 - Memory
- ❑ Azure SRE Agent – Introduction
- ❑ Characteristics of Azure SRE Agent – Explained
- ❑ DEMO



AGENTIC AI

Agentic AI refers to a new generation of artificial intelligence systems that act autonomously to achieve predetermined goals with minimal or no human oversight

Scenario: Email Generation Using an LLM

Prompt: "Write an email thanking a client for a meeting and attach a proposal."

LLM Response:

Email  

Subject Thank You for the Meeting and Proposal Submission

Dear [Client Name],

Thank you for taking the time to meet with us today. It was a pleasure discussing your requirements and understanding your expectations in more detail.

As discussed, please find the attached proposal for your review. The document outlines our approach, scope, and next steps. We believe this aligns well with your objectives and would be happy to walk you through any section or address any questions you may have.

Thank you once again for the opportunity. I look forward to your feedback and to working together.

Best regards,
Rohit K Singh



Agent Workflow:

Example: Weather Agent

User Query: "How is the weather in New Delhi?"



➤ LLM does the analysis, if tools are required?

➤ It extracts the relevant entity (e.g., "New Delhi.")

➤ The request is sent to the weather API (tool)

➤ The API returns data, which LLM uses as context

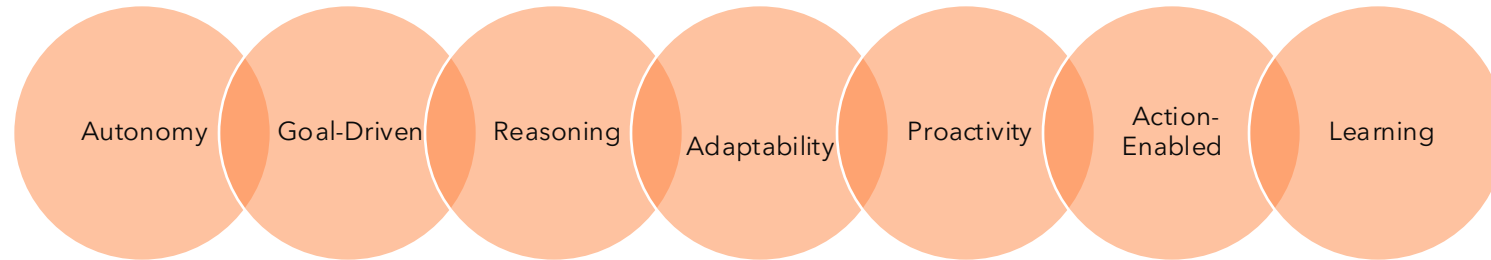
➤ The agent delivers the response

Response: "The weather in New Delhi is 32°C."



AGENTIC AI

❑ Key characteristics:



Ex: "Set up a CI/CD pipeline," and it creates, configures, and runs it without asking you at every step

Ex: If the goal is "reduce cloud cost," it analyzes usage and suggests or applies optimizations

Ex: if deployment fails, it checks logs and decides whether to retry or fix the issue

Ex: If one deployment method fails repeatedly, it switches to a more reliable strategy

Ex: Detects high CPU usage and scales servers before the system crashes

Ex: Instead of telling you how to deploy, it runs commands and deploys the app

Ex: After multiple deployments, it learns the fastest and safest way to release updates



AGENTIC AI

❑ AI Agent Framework:

LangGraph

Crew AI

Auto Gen

Google
(ADK)

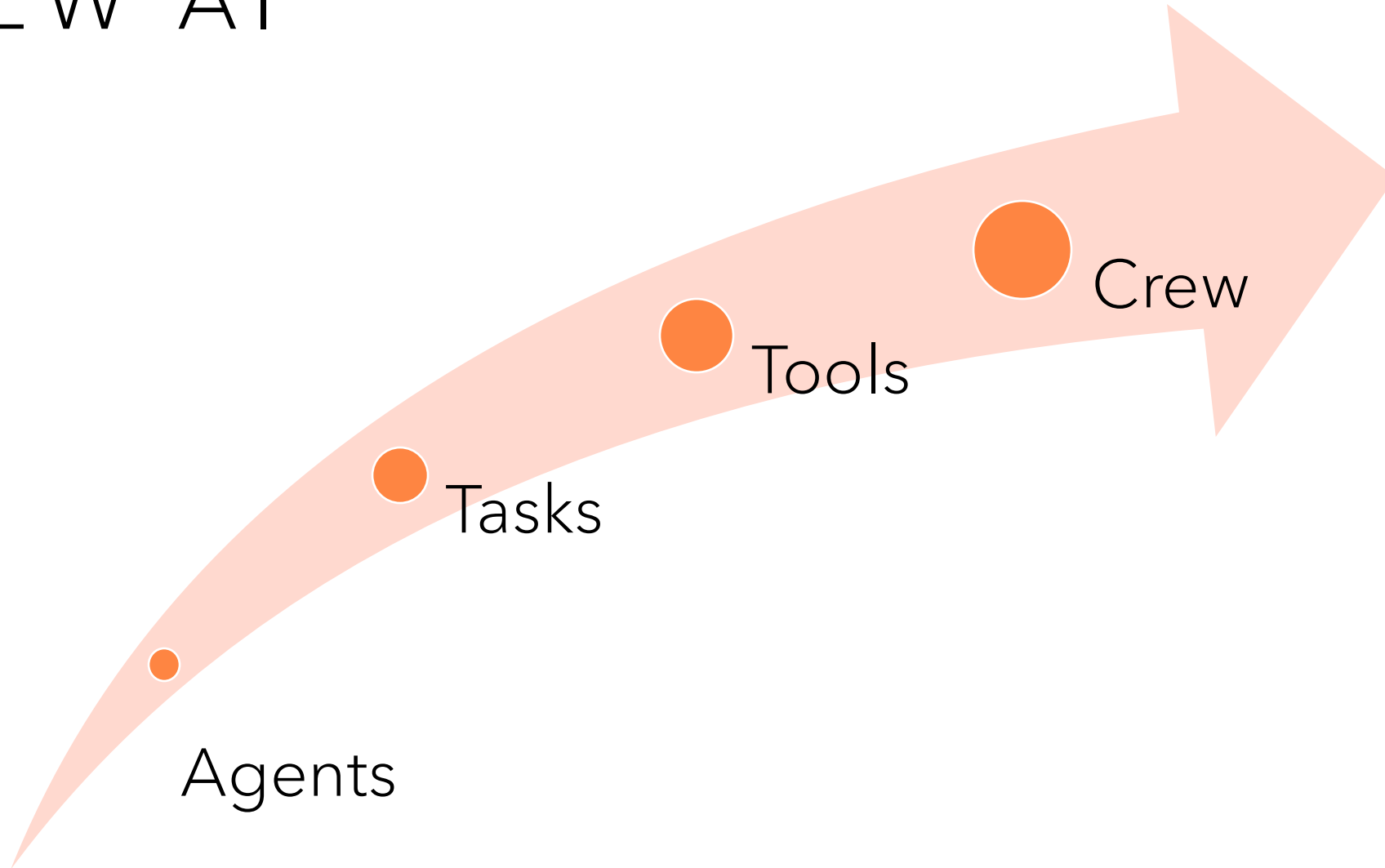
HuggingFace



CREW AI

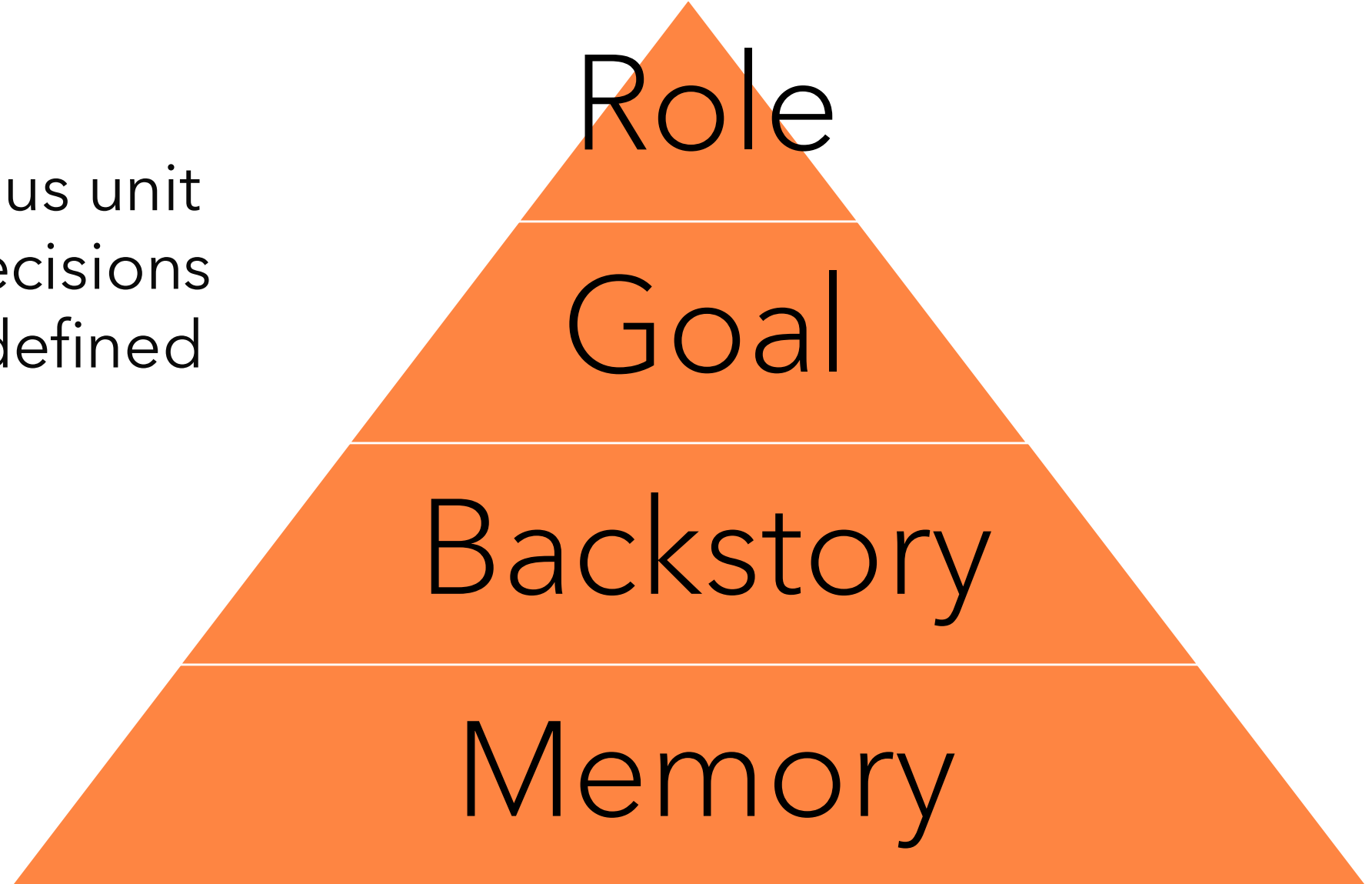
An open-source Python framework designed to orchestrate **role-playing autonomous AI agents** to complete complex tasks

CREW AI



CREW AI AGENTS

An autonomous unit
that makes decisions
based on its defined
identity

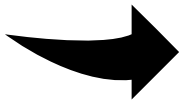


AZURE SRE AGENTS



AZURE SRE AGENT

❑ An AI-powered reliability service that continuously monitors Azure resources

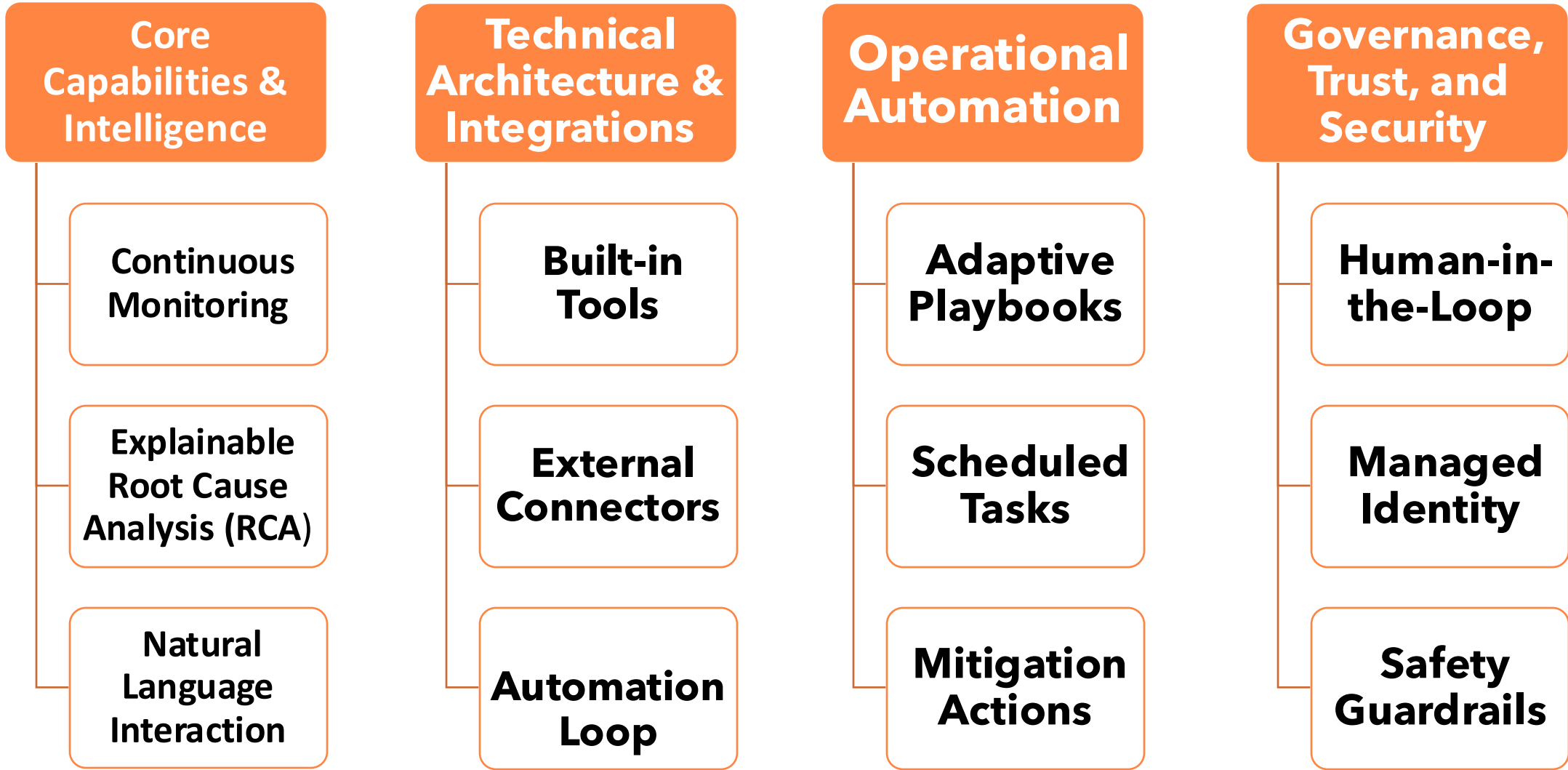


To reduce manual "toil" and improve system uptime through automated diagnosis and mitigation



Provides an "always-on" SRE assistant that understands your telemetry, runbooks, and incident tools

- Monitors system
- Detect Issues
- Fix problems automatically
- Improve reliability
- Incident handling



DEMO

MODEL CONTEXT PROTOCOL (MCP)



AGENDA

- ❑ Model Context Protocol – Introduction
- ❑ MCP Workflow – Explained
- ❑ DEMO – Setting UP 1st MCP Server Locally



HTTP/HTTPS

TCP/IP

WHAT IS PROTOCOL?

SMTP

Wi-Fi

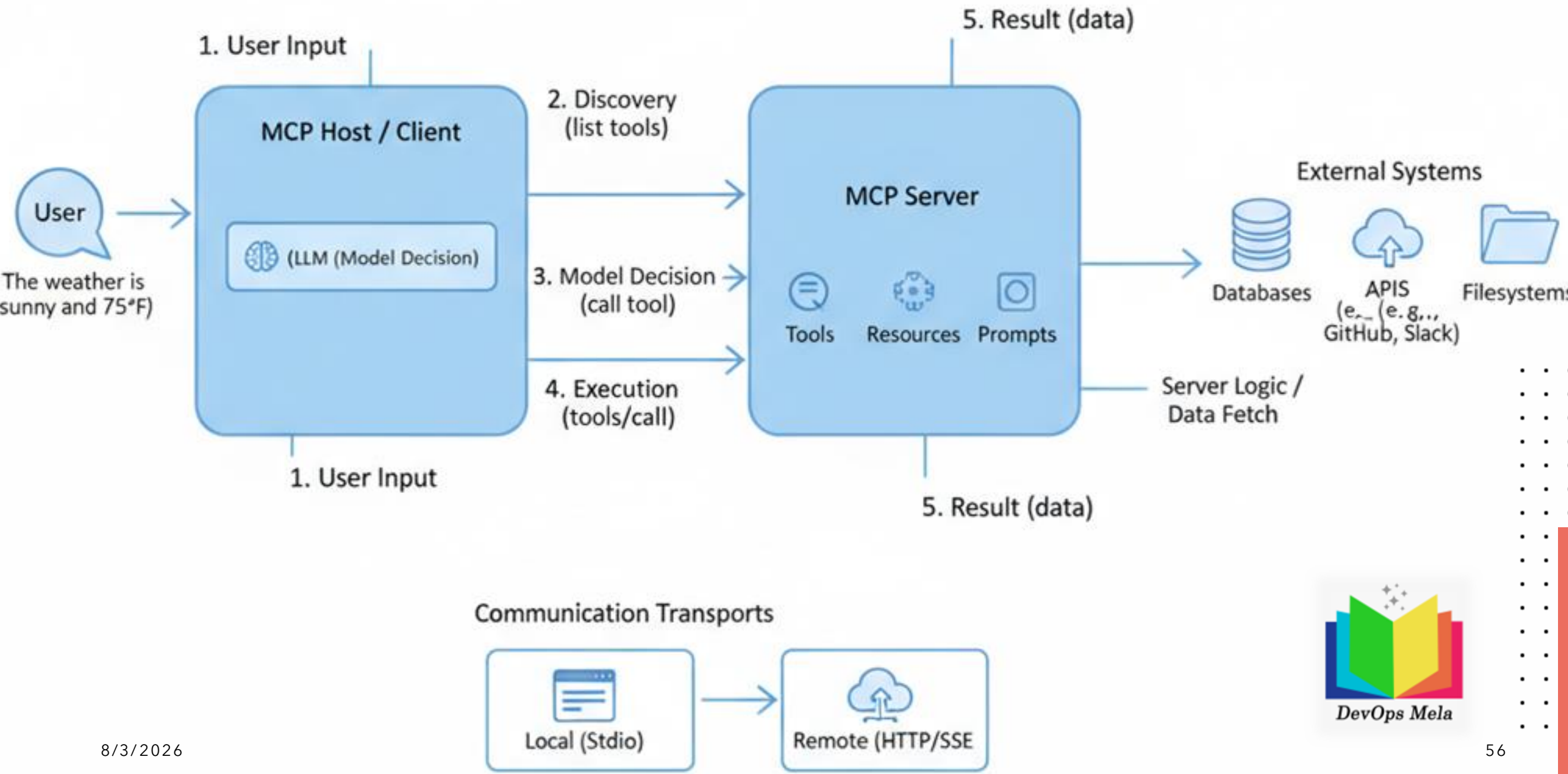


MODEL CONTEXT PROTOCOL (MCP)

- ❑ MCP, introduced by Anthropic in late 2024, serves as the "USB-C" for AI integrations—a universal, standardized protocol that simplifies and streamlines these connections



How it MCP works?



GITHUB MCP SERVER



AGENDA

- ❑ GITHUB – Introduction
- ❑ Understanding basic concepts of GITHUB
- ❑ GITHUB MCP – Introduction
- ❑ Various Use-Cases of GITHUB MCP – Explained
- ❑ DEMO



GITHUB ACTION

```
name: hello-world!!
```

```
on: push
```

```
jobs:  
  simple-hello-world:
```

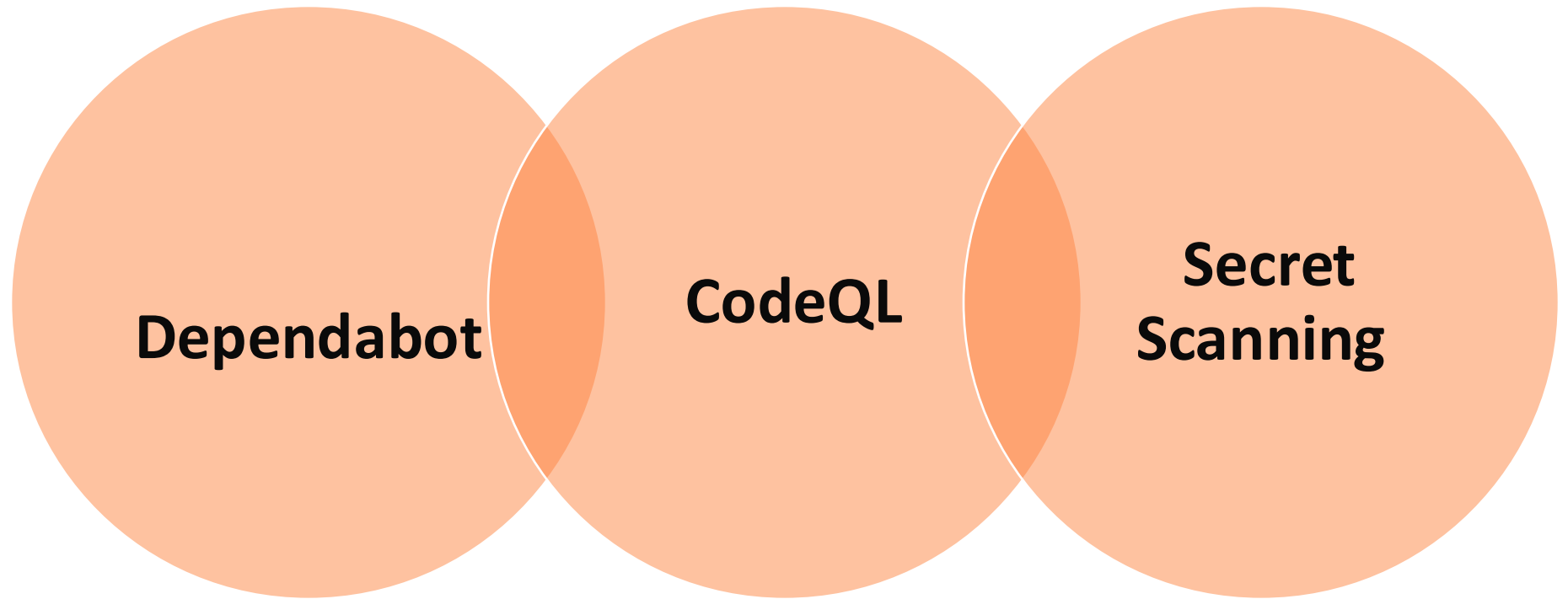
```
runs-on: ubuntu-latest
```

```
steps:  
  - name: Checkout Repo  
    uses: actions/checkout@v6
```

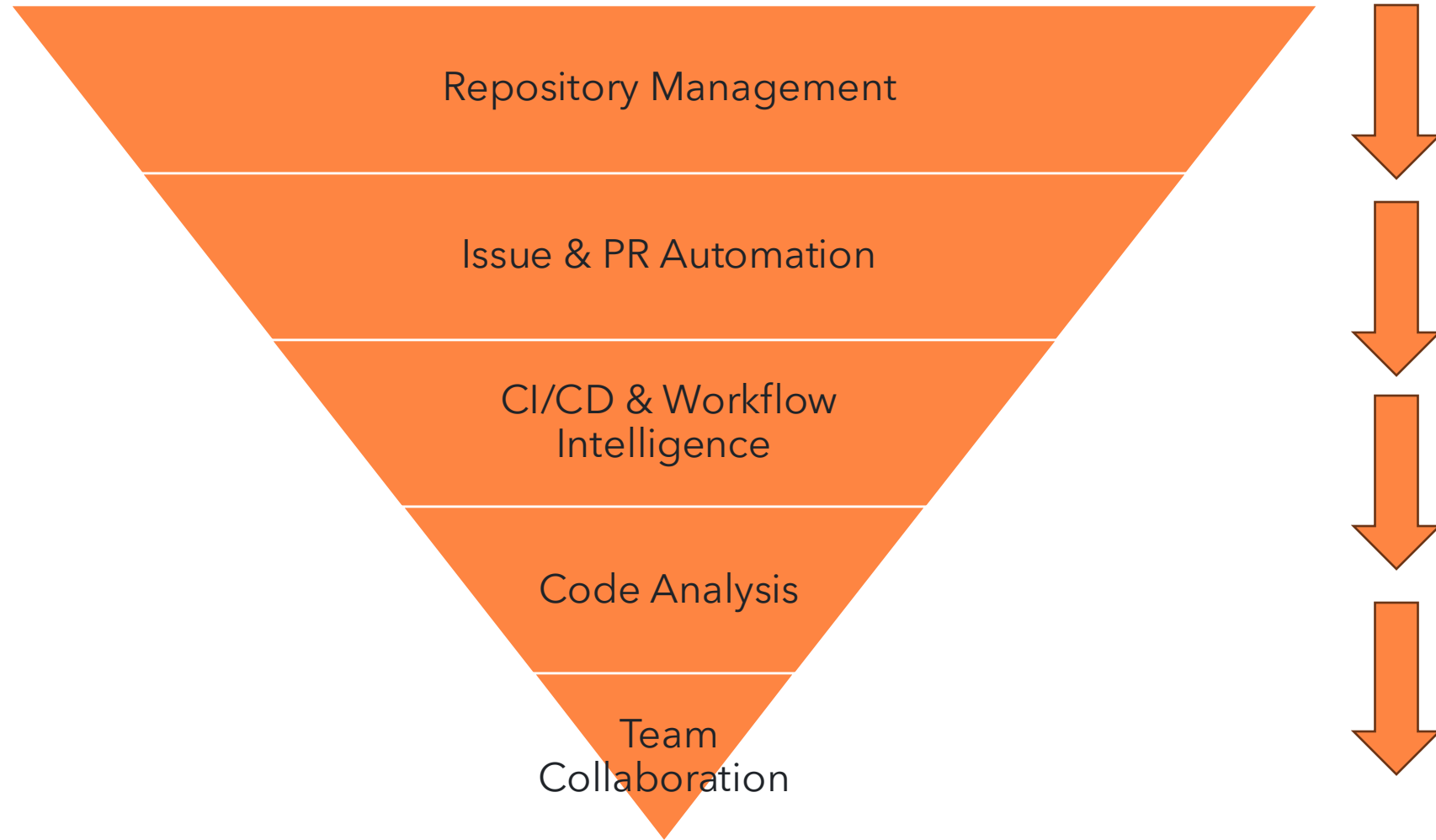
```
- run: echo "$(cat hello_world.txt)"
```



Security Features



USE CASE'S



DEMO

AZURE DEVOPS MCP SERVER



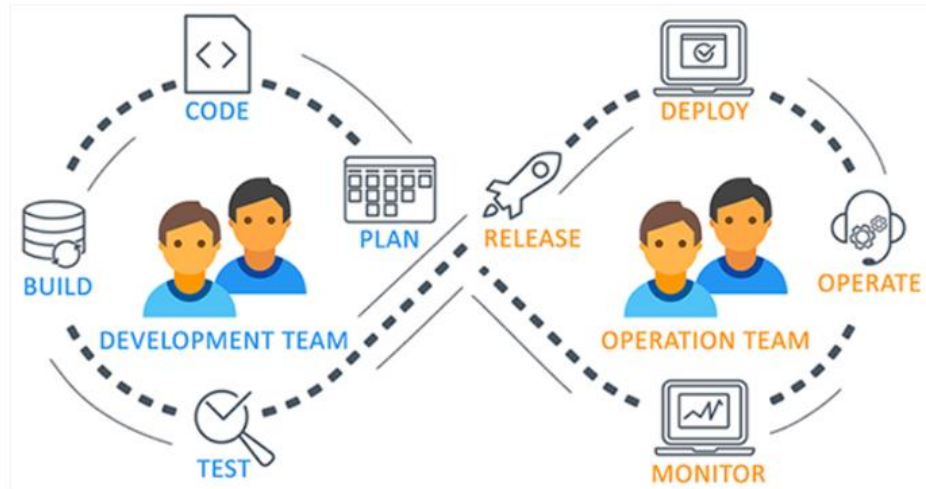
AGENDA

- ❑ Azure DevOps – Introduction
- ❑ Understanding basic concepts of Azure DevOps
- ❑ Azure DevOps MCP – Introduction
- ❑ DEMO



Azure DevOps:

- DevOps is the union of people, processes, and products to enable continuous delivery of value to the end users
- Using Azure DevOps, we can plan smarter, collaborate better, and deliver reliable software faster with a set of modern services



☐ Azure Boards

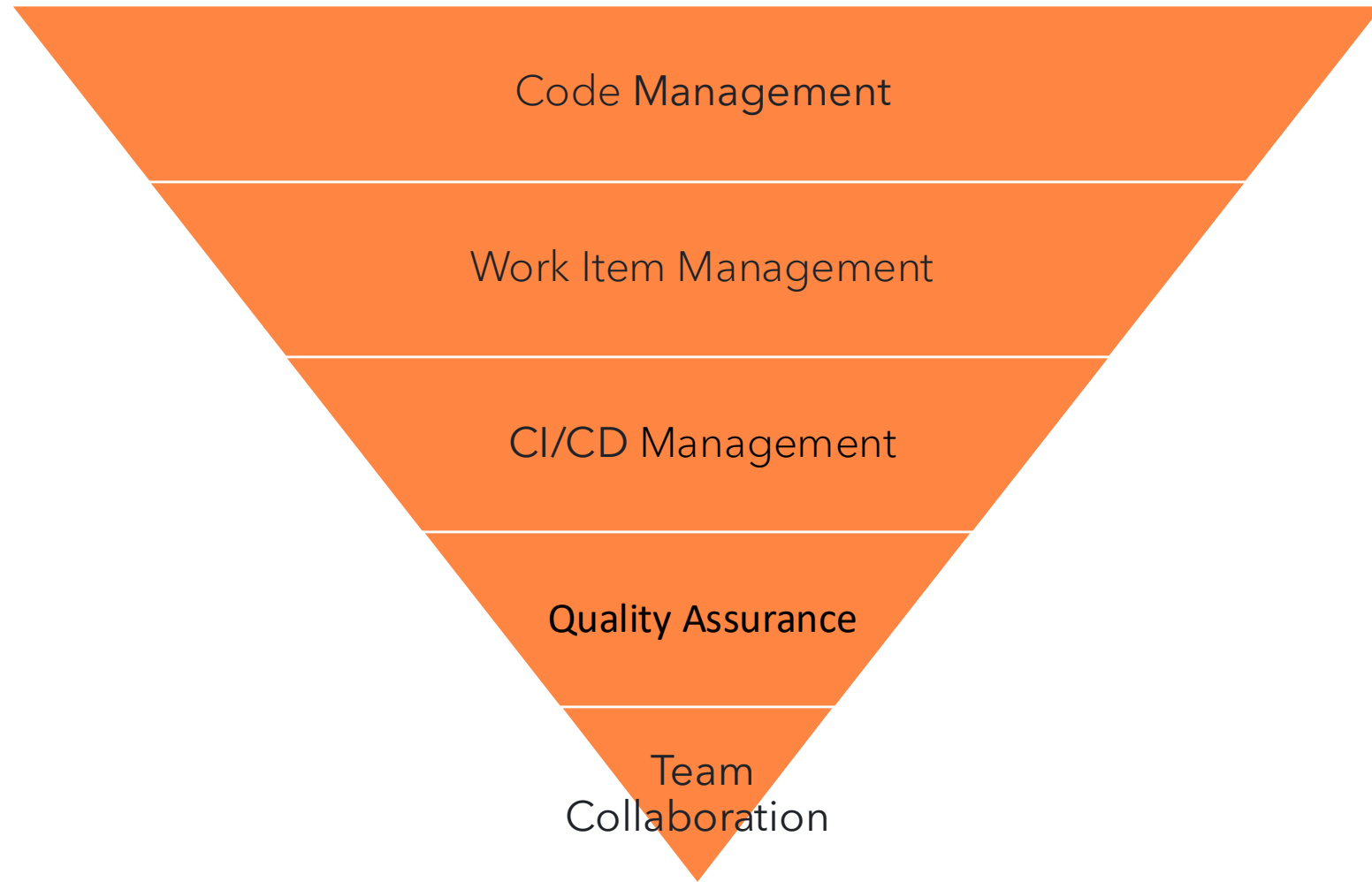
☐ Azure Test Plans

☐ Azure Pipelines

☐ Azure Repos

☐ Azure Artifacts

USE CASE'S



DEMO

TERRAFORM MCP SERVER



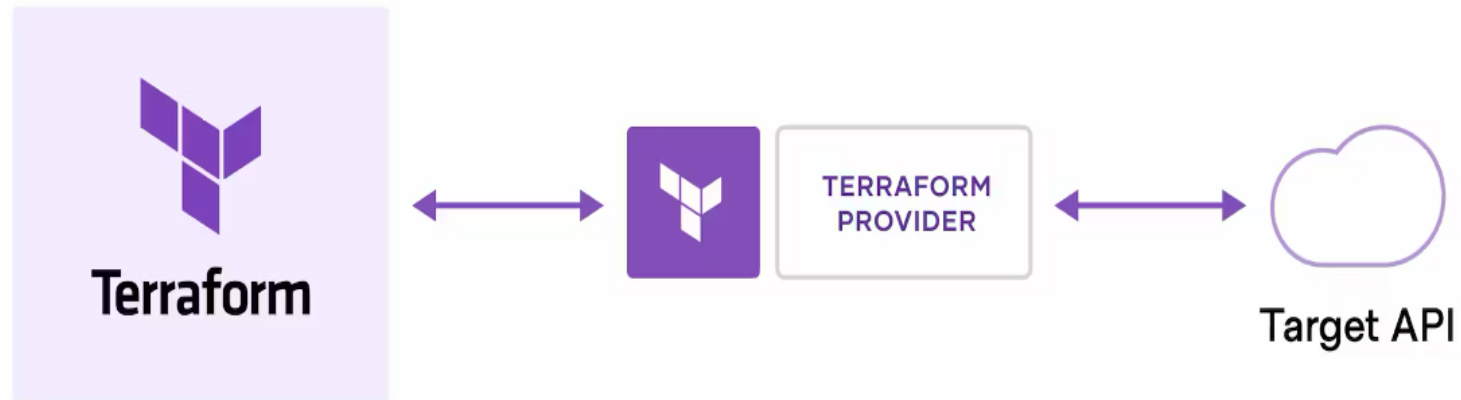
AGENDA

- ❑ Terraform - Introduction
- ❑ Setting up Terraform - Locally
- ❑ Understanding Terraform Workflow
- ❑ Terraform State file and Providers - Explained
- ❑ Terraform MCP - Introduction
- ❑ Various Use-Cases of Terraform MCP - Explained
- ❑ DEMO

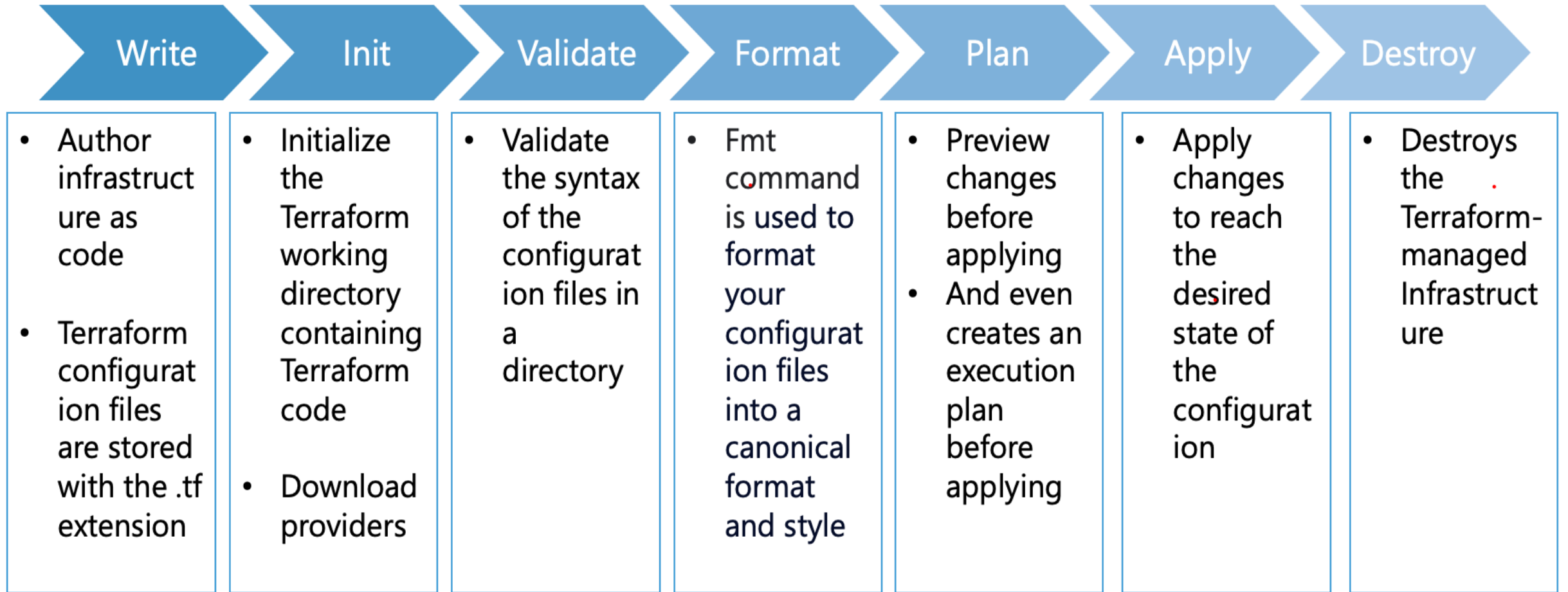


Terraform:

- HashiCorp Terraform is an **infrastructure as code** tool that lets you define both cloud and on-prem resources in human-readable configuration files that you can version reuse, and share
- Terraform work-flow we can easily provision and manage all of your infrastructures throughout its lifecycle
- Terraform we can manage low-level components like computing, storage, and networking resources, as well as high-level components like DNS entries and SaaS features



Workflow:



Terraform State File:

- Terraform stores information about your infrastructure in a state file
- This state file keeps track of resources created by your configuration and maps them to real-world resources

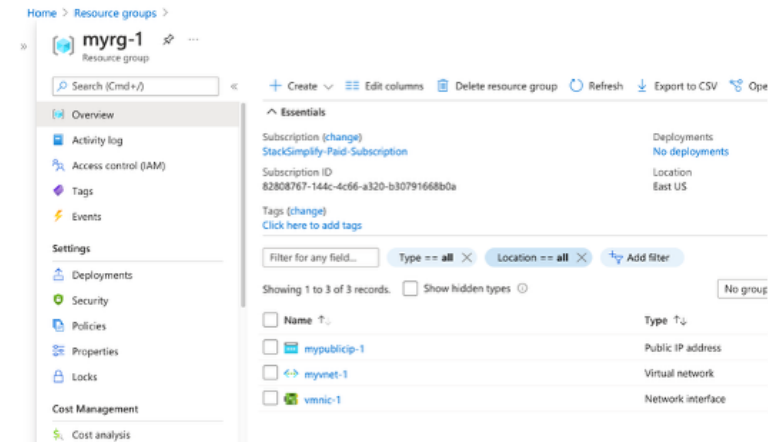
Terraform Configuration Files

```
06_Terraform_Provider
├── terraform-manifests
│   ├── c1-versions.tf
│   ├── c2-resource-group.tf
│   └── terraform-manifests_2
```

Desired State

Terraform
State file

Real World Resources



Home > Resource groups > myrg-1

Search (Cmd+J)

Overview

- Activity log
- Access control (IAM)
- Tags
- Events

Settings

- Deployments
- Security
- Policies
- Properties
- Locks
- Cost Management
- Cost analysis

Essentials

Subscription (change)
StackSimply-Paid-Subscription

Subscription ID
82808767-144c-4c66-a320-b30791668b0a

Location
East US

Tags (change)
[Click here to add tags](#)

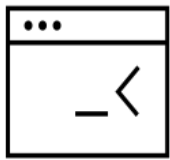
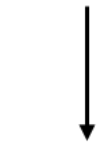
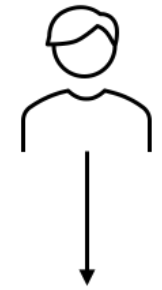
Filter for any field... Type == all Location == all Add filter

Showing 1 to 3 of 3 records. Show hidden types

Name	Type
mypublicip-1	Public IP address
myvnet-1	Virtual network
vmnic-1	Network interface

Current State

Terraform Providers:



Download Provider

1 Terraform Init

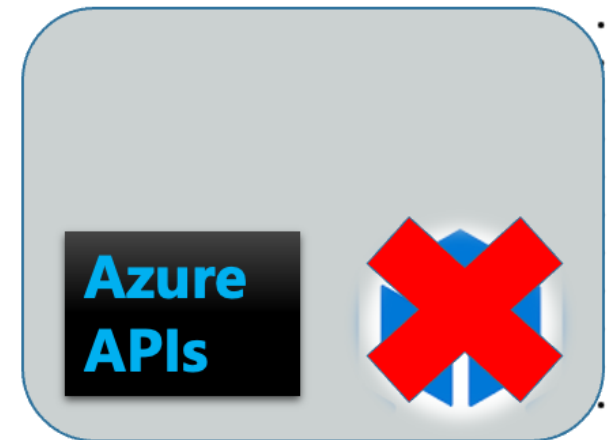


3 Terraform Plan

4 Terraform Apply

2 Terraform Validate

5 Terraform Destroy



Provider is a piece of software that implements Terraforms resource type definitions

Implements the CRUD (Create, Read, Update, Delete) operations for those resource types

Providers are an important component to manage resources

Each provider has its own release cycle and gets versions separately

Terraform registry maintained and stored providers for each major and minor cloud infrastructure providers

USE CASE'S

**Interactive
Documentation
& Discovery**

**Guided
Configuration
Generation**

**Error
Resolution and
Troubleshooting**

**Infrastructure
"What-If"
Analysis**



Comparison: Standard AI vs MCP Enabled AI

Feature	Standard AI (GPT-4 / Claude 3)	MCP-Enabled AI
Knowledge Cutoff	May use outdated provider versions	Fetches real-time Registry data
Accuracy	Often "hallucinates" resource attributes	Queries official schema definitions
Integration	Requires copy-pasting docs	Direct access to Registry/Modules
Focus	General coding assistance	Specialized Infrastructure as Code (IaC) support

OLLAMA

AGENDA

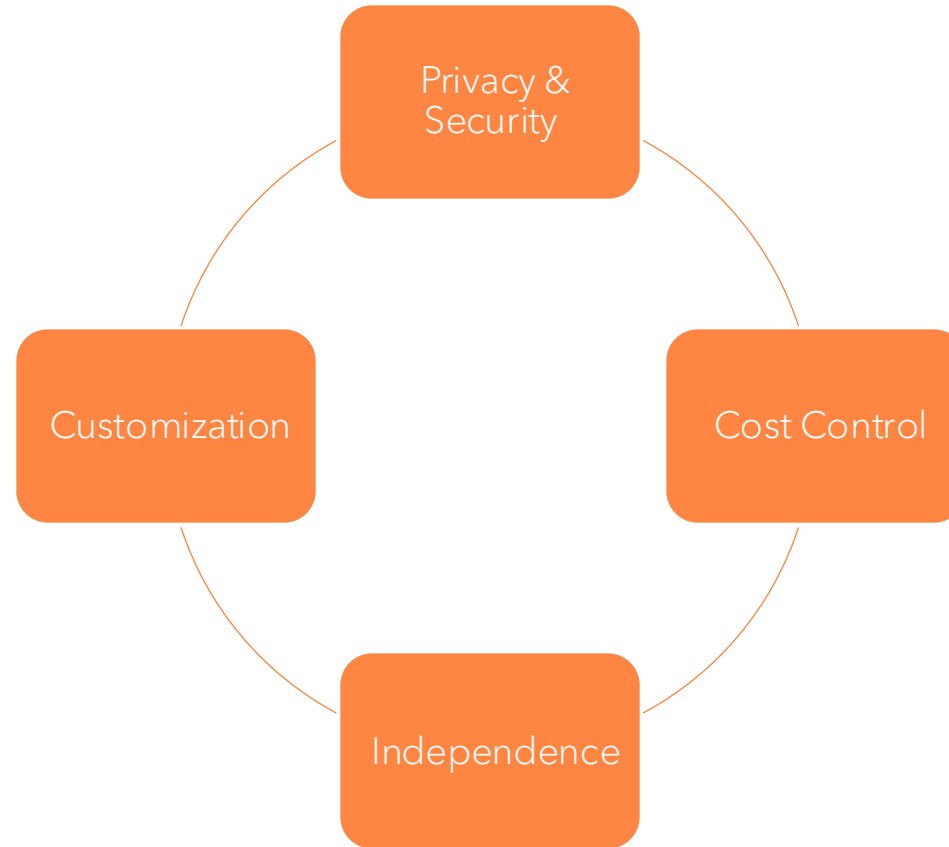
- ❑ OLLAMA - Introduction
- ❑ Setting up OLLAMA - Locally
- ❑ DEMO



What is OLLAMA?

- ❑ A lightweight, extensible framework designed to run Large Language Models (LLMs) locally on your own machine

Why Run LLMs Locally?



DEMO

DEMO IDEA:

- Setting up SRE Agent and diagnosing 1st incident
- Starter Prompts - Azure Services & AKS
- Creating .NET Core with Application Insights integration
 - Creating ALERTS
 - Setting up incident plan in SRE Agents to auto-mitigate the issue

DIFFERENCE

Generative AI

- Broader category encompassing various content generation types, including text, images, music, and more
- Can produce diverse content types such as text, images, music, and more
- Uses diverse datasets depending on the purpose, like images for image generation or text for text generation
- Wider range of applications, including creative industries, entertainment, content generation, and more

Large Language Models (LLMs)

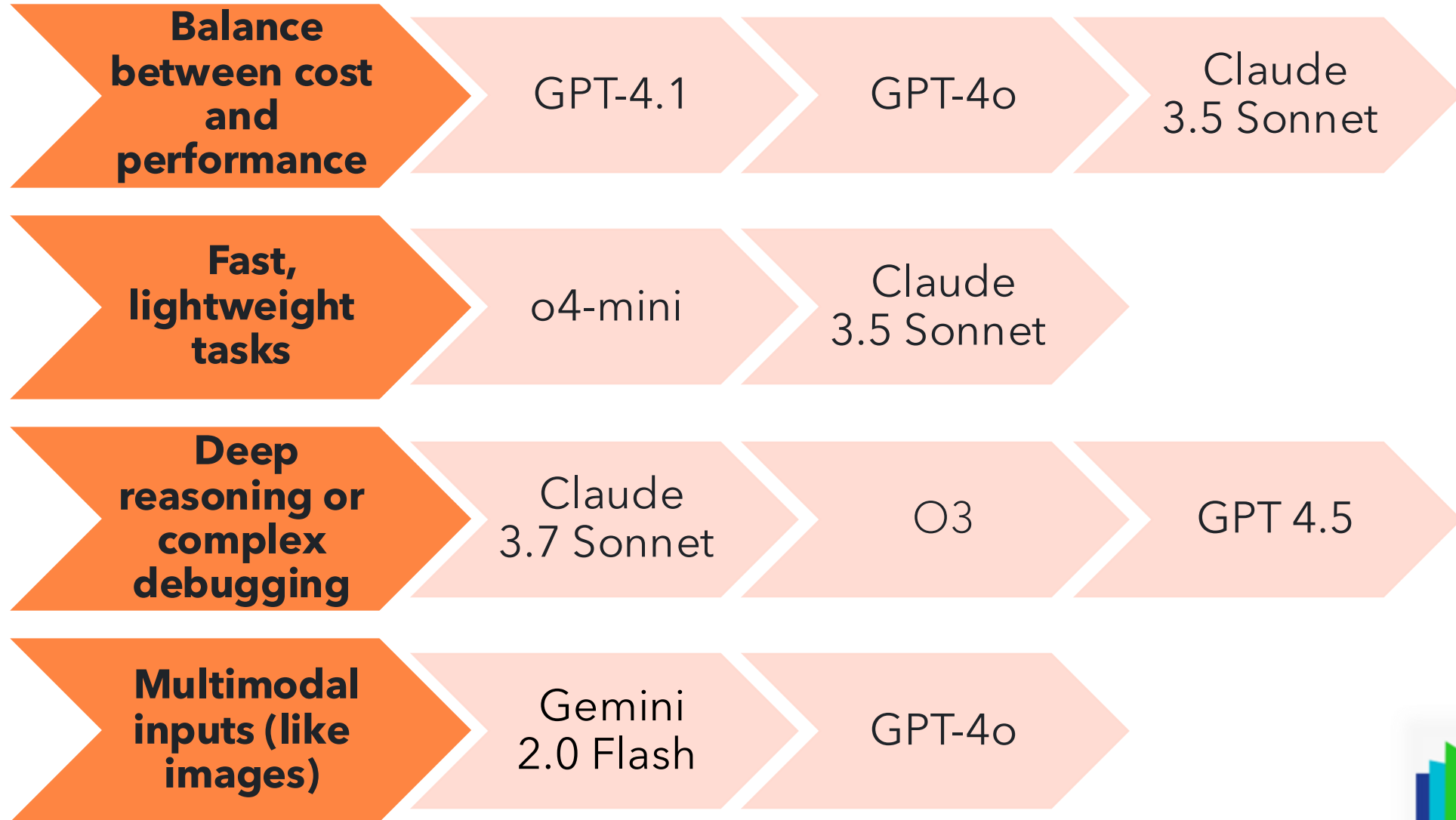
- A subset of generative AI specifically focused on language-related tasks
- Primarily generate text-based outputs
- Trained exclusively on text data
- Particularly suited for language-related tasks and natural language processing like chatbots and summarization



GEN AI - MODELS

Models	Description	How they works	Use-cases	Example
Large Language Models (LLMs)	LLMs are specialized in understanding and generating human-like text by processing extensive amounts of language data	LLMs use a transformer architecture and deep learning techniques to understand language context, predict the next word in a sequence, and generate coherent text	Content creation, Chatbots and virtual assistants, Translation and summarization, Code generation Data analysis	ChatGPT (OpenAI) Gemini (Google DeepMind) Claude (Anthropic)
Generative Adversarial Networks (GANs)	GANs involve a two-network system: a generator that creates synthetic data, and a discriminator that assesses its authenticity	The generator strives to create data indistinguishable from real data, while the discriminator strives to differentiate between real and generated data, driving both networks to improve	Image and video synthesis, Style transfer, Medical imaging, Data augmentation	StyleGAN (NVIDIA) CycleGAN Deep Convolutional GAN (DCGAN)
Diffusion models	Diffusion models work by systematically introducing noise to training data and learning to reverse this process to reconstruct the original data or generate new outputs from random noise	These models learn the underlying patterns and structures of data by adding noise, then iteratively refine random noise to generate realistic output	High-quality image and audio generation, Image enhancement and restoration, Scientific research	DALL-E 3 (OpenAI) Stable Diffusion
Variational Autoencoders (VAEs)	VAEs learn a compressed representation (latent space) of the input data and generate new data samples by sampling from this learned distribution	An encoder compresses input data into a lower-dimensional latent space, and a decoder reconstructs the input from this representation. VAEs learn the underlying structure of the data and generate new samples based on that structure	Data compression and generation, Anomaly detection, Image generation	CompressionVAE PyTorch Examples (VAE)

WHICH MODELS TO USE



RAG

QWEN AI INTEGRATION



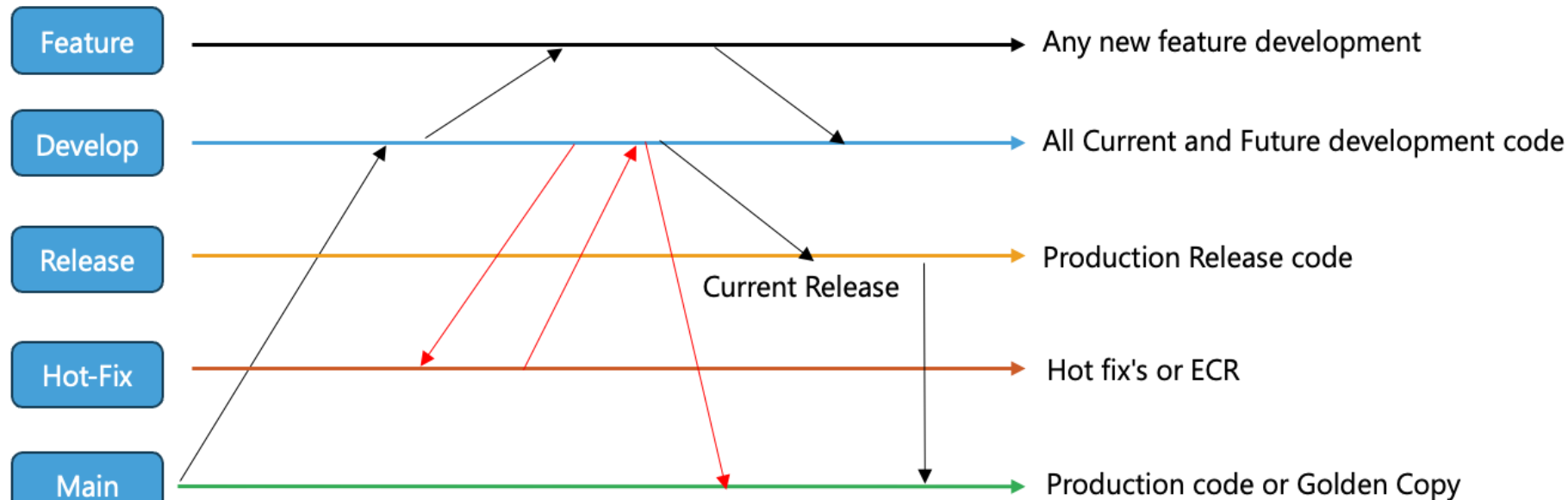
Branching Strategies:

- Effective branch management is crucial for successful collaboration and efficient development with Git

Git-Flow is a comprehensive branching strategy that aims to cover various scenarios
It defines specific branch responsibilities, such as

□ Git-Flow:

- main/master for production
- develop for active development
- feature for new features
- release as a gatekeeper to production
- hotfix for addressing urgent issues



Branching Strategies:

❑ Trunk-based:

Trunk-based development is a version control management practice where developers merge small, frequent updates to a core “trunk” or main branch

